

DSC 399: Advanced Applications and Interpretability of Neural Networks

Syllabus

Instructor Information

- **Instructor Name:** Dr. Julie Butler
- **Instructor Email:** butlerju@mountunion.edu
- **Instructor Cell Phone:** 864-993-7133
- **Class Meeting Days and Time:** MWF 8:45am - 9:50am
- **Class Meeting Location:** Tolerton Hood Hall Room 102
- **Final Exam Date and Time:** Friday March 6th, 10:00am - 12:00pm
- **Instructor Office Hours:** Monday/Wednesday 2:00pm - 4:00pm, Thursday 8:30am - 10:00am, Tolerton Hood Hall 102

Course Description

This course examines advanced applications of neural networks within fields such as natural language processing, computer vision, reinforcement learning, and generative modeling. Students will explore modern neural network architectures, including attention mechanisms, transformers, graph neural networks, and deep generative models. This course will also consider the interpretability and scalability of these modern architectures, along with the ethical considerations of gathering data to train these models.

Course Purpose

This course serves as a follow-up for DSC 340 and will cover applications of neural networks which are important in the modern world. These applications will also blend machine learning and artificial intelligence, making them increasingly important in modern computing landscape.

Prerequisites

DSC 340 or permission of the instructor.

COURSE LEARNING OUTCOMES:

- Understand and implement modern neural network architectures with application to text analysis, computer vision, graph analysis, and generative modeling.
- Evaluate models in terms of interpretability and scalability.
- Discuss real-world applications of the modern neural network architectures along with ethical considerations that come from these applications

TECHNOLOGY REQUIREMENTS:

College coursework requires students to be responsible with reading and assignments, checking email and D2L frequently, and staying in regular communication with instructors. Technology access will be important for success. To participate in learning activities and complete assignments, you will need: * Access to a working computer that has a current operating system with updates installed * Reliable Internet access and a Mount Union email account * A current Internet browser that is compatible with D2L * A current version of Python and Jupyter notebooks installed with the following libraries: Numpy, Pandas, SciPy, Seaborn, Scikit-Learn, Tensorflow, PyTorch. * As an alternative to running programs locally you are allowed to use Google Colab, a cloud-based Jupyter notebooks installation from Google.

Please contact the IT Help Desk at 330.829.8726 or helpdesk@mountunion.edu if you need assistance with obtaining or using a device, any necessary software, or internet access at any time during this semester .

COURSE MATERIALS – RAIDER READY PROGRAM:

This course is part of the campus' course material delivery program, Raider Ready. The bookstore will provide each student with a convenient package for physical books and any digital materials have been integrated into the D2L page for this course. You should have received an email from the bookstore to confirm your materials. If you have not completed that confirmation, please do so as soon as possible. Be sure to then pick up your physical materials in the bookstore. Make sure to return all textbooks to the University Store BEFORE you leave for break!

For more information on Raider Ready, please go to mountunion.edu/raider-ready. Or Visit the University Store

All course materials comply with copyright/fair use policies.

Course Textbook: *Artificial Intelligence: A Modern Approach*. Stuart Russell and Peter Norvig. 4th ed.

Grading

- **Quizzes (4): 30%**
 - This score is weighted using the following formula, $0.4A + 0.3B + 0.2C + 0.1D$, where A is the highest score, B is the next highest score, C is the third highest score, and D is the lowest score.
- **Homework (4): 20%**
 - This score is weighted using the following formula, $0.4A + 0.3B + 0.2C + 0.1D$, where A is the highest score, B is the next highest score, C is the third highest score, and D is the lowest score.
- **Attendance and Participation: 20%**
- **Final Project: 30%**

Grading Scale

- A: 94% - 100%
- A-: 90% - 93%
- B+: 87% - 89%
- B : 84% - 86%
- B-: 80% - 83%
- C+: 77% - 79%
- C: 74% - 76%
- C-: 70% - 73%
- D+: 67% - 69%
- D: 64% - 66%
- D-: 60% - 63%
- F: 59% and below

Quizzes

Four quizzes will be given throughout the semester, testing your knowledge of the concepts and algorithms taught in lecture. Students will be given 30 minutes to complete each quiz, which will contain 5-10 short answer questions. A practice quiz will be provided to students at least one week prior to each quiz.

Homework

Four homeworks will be given throughout the course, each one furthering your knowledge of the algorithms, code implementations, and concepts. For questions that require a coding implementation, correct and documented code must be provided for full credit. For questions which require a text explanation, the explanations must be provided in the homework notebook in formatted markdown cells. Expect to spend approximately 15 - 20 hours on each homework assignment over the course of a two week period. Note that these assignments will have long run times.

Attendance and Participation

As this is a flipped classroom, your attendance and participation in all class meetings is required. For each class meeting you will receive an attendance score out of 5, with points being deducted for arriving late and no points received for an unexcused absence. An excused absence requires a note from the health center or another health care provider, a note from a coach in the case of an athletic absence, or a note from a professor in the case of a field trip. In the case of a family emergency or other situation that will require an absence, clear communication prior to the absence is required. Any unexcused absences will receive an attendance score of 0. For each class meeting you will also receive a participation score out of 5, with full points being awarded for steadily working on the week's projects. Points will be deducted for phone use, off topic conversations, etc.

Final Project

The final project will allow you to apply concepts from the course to a problem that is unique to your interest. The final project must use machine learning to solve a problem, with some restrictions. If you plan to use an algorithm that was covered either in this course or in DSC 340, then you must apply the algorithm to a type of data which we have not covered (for example applying a recurrent neural network to analyze sound data). If you plan to use a type of data that we have used in this course or in DSC 340, then you must use an algorithm that was not covered in either course. Each student must give a 15 minute presentation on their work during the final exam period for the course. More details of the final project can be found on D2L.

COURSE POLICIES/PROCEDURES/EXPECTATIONS

COURSE FLEXIBILITY:

This course, syllabus, and assignments are subject to change at the discretion of the instructor.

COURSE COMMUNICATION:

Students are expected to check their Mount Union email account frequently for important course information. Any student having trouble understanding any aspect of the course should reach out to the instructor. The best method is to email or to attend office hours. If a quick response is required, then you may text me using the phone number at the beginning of this document.

ELLUCIAN ADVISE:

Advise is a communication and early alert software designed to support and encourage student success. Faculty and staff may use this system to send notices to students. Keep an eye out for messages. If and when a message is received, take action – talk to a professor, an advisor, or someone in the Center for Student Success to get on the path to success!

LAPTOP/SMARTPHONE POLICY:

All electronic devices are allowed in the classroom, if you do not use them to distract yourself or other students. All devices should be muted and notifications silenced for the class duration. If a device distracts other students, you will be asked to put the device away or leave the classroom.

LATE WORK POLICY:

No late work is accepted in this course for any assignment except in special circumstances (family emergency, extended illness, etc.). If you are experiencing a special circumstance and would like to request an extension on an assignment, **you must contact the instructor at least six hours before the deadline for an extension.**** Extensions will be granted at the instructor's discretion and are not guaranteed.

ACADEMIC INTEGRITY:

Academic Integrity is at the heart of the mission and values of the University and is an expectation of all students. Maintaining academic integrity is a reflection of character and personal integrity and is a means to achieving the outcomes of this course and that grades accurately reflect learning and understanding of the course material. Mount Union expects all students to be responsible members of the academic community, so all work submitted in this course should be yours and yours alone. Please refer to the Academic Honesty section in the current Undergraduate Catalog and Student Handbook.

All work you submit with your name on it is expected to be original work. You can consult any outside source, including the internet and AI chats, for help on the class projects, but you are not allowed to copy any solutions you find there directly. Additionally, you should be able to thoroughly explain how you arrived at your answer for all work you turn in; all assignments are subject to possible verbal discussion which can contribute to your grade. If you work closely with other classmates on an assignment, please indicate that the solution results from collaboration and list the names of all students who contributed (this is allowed and encouraged). If it can be proven that you copied directly from Chegg, ChatGTP, or another person on your work (i.e., you are copying your solutions directly from these sources or others), you will receive a zero for the assignment and be reported for academic dishonesty.

UNIVERSITY AI POLICY

The university defines misuse of AI as using AI without the permission of your instructor. Some potential examples of misuses of AI are: * Presenting AI-created works as your own. * Using AI to produce, revise, or substantially edit written or creative work. * Solving equations or creating code with AI. * Using AI to produce speech outlines or slide decks. * Any other use that is not expressly permitted by your instructor.

If you are unsure about whether you may be violating the university policy, please check with your instructor before turning in your work.

CLASSROOM AND ONLINE CONDUCT:

Professionalism will be expected at all times. Because the classroom is a place designed for the free exchange of ideas, respect for one another is expected in all circumstances. This includes both in-person and online forums. Appropriate language and restraint from verbal attacks upon those with differing perspectives is a minimum requirement. Bullying will not be tolerated.

STUDENT RIGHTS AND GRIEVANCES:

Students have a right to equitable access for their course instruction, materials, and facility. Students also have a right to respectfully express their views and opinions in class discussions (both in-person and online), as well as in their writing and speaking assignments, without fear of ridicule or retribution. In addition, students have a right to safety and privacy during their course instruction that prohibits discrimination and harassment of any type by any person. Students also have a right to fairness in grading and learning outcome assessment, as well as timely communication from the instructor. Any student who may feel that their rights have been violated may file a grievance regarding Title IX, civil rights, conduct, discrimination, or harassment using the appropriate grievance procedure in the University's student handbook available on the Mount Union website. Academic complaints, such as disputes over a grade,

should be filed in accordance with the procedures laid out in the University Catalogue available on the Mount Union website.

CAMPUS RESOURCES FOR STUDENT SUCCESS

CENTER FOR STUDENT SUCCESS:

The Center for Student Success provides numerous support services for students on campus such as, tutoring, getting organized, developing better study techniques, preparing for internships, getting ready for a career and/or graduate school, or making choices related to the major. To make an appointment for any of these services, visit the Center for Student Success in the lower level of the library (KHIC) or reach out to make an appointment (studentsuccess@mountunion.edu or 330-829-8107).

INTEGRATIVE CORE OFFICE:

The Integrative Core Office (IC Office) helps with FYS courses, Foundations (HANS) courses, the second-year Raider Foundations Portfolio (RFP), Explorations (VG) courses, and IC Capstone courses. The IC Office has an open-door policy that does not require appointments for individual consultations, but both in-person and virtual appointments are available. Stop in KHIC 110A, call 330.829.8229, or email icore@mountunion.edu.

DWOC STUDIO:

The Digital, Written and Oral Communication (DWOC) Studio is located in the Mount Union Library, across from the circulation desk. Its mission is to provide space, technologies, and peer consultations to Mount Union students who are working on writing, oral presentations, and multimedia projects, including the Raider Foundations Portfolio (RFP). Faceto-face and synchronous online consultations are available to all undergraduate students. See the DWOC Studio website for more details.

MATHEMATICS LEARNING CENTER:

Located on the M level of KHIC, the Mathematics Learning Center (MLC) is in place to work with all UMU students in need of assistance in traditional math courses, as well as courses with math-related material such as accounting, chemistry, economics – anything that requires an understanding of math, formulas, or theorems. The MLC has daily walk-in availability most afternoons and evenings as well as the ability to set up appointments with the director. Stop by KHIC 258, email mlc@mountunion.edu, or call 330.829.6643.

LIBRARY SERVICES:

Located in KHIC and Cope Music Hall, the Mount Union Libraries offer assistance to students at any stage in the research process. In addition to providing access to a wide range of books, ebooks, and scholarly articles, the librarians also offer their expertise in finding information to help students with their research project, no matter the discipline or topic. Whether it be a quick question about finding that last elusive source for a paper, scheduling a longer consultation to go over their projects in greater depth or learn the fundamentals of researching, or anything in-between, the reference librarians are there to make students' lives a little easier. Students also have access to OhioLINK, a statewide library system that greatly expands the amount of materials students can obtain at no cost. Library staff welcome the opportunity to collaborate with students on research assignments. Visit the reference desk on weekdays, check out our [FAQs](#), or [submit a question for assistance](#).

ACCESSIBILITY STATEMENT:

Student Accessibility Services (SAS), located in the Center for Student Success, in the lower level of Kolenbrander-Harter Information Center (KHIC), coordinates services and accommodations for students with documented disabilities, diagnosis, and/or health-conditions. To initiate services, students should meet with the Director of Student Accessibility Services as soon as possible to discuss reasonable accommodation options. A student needs to formally request and be approved for accommodations in order for the faculty member to provide accommodations in the classroom. Accommodations are not retroactive, so students are encouraged to make requests as soon as possible. You may contact the Director of Student Accessibility Services by phone at (330)-823-7372; or by e-mail at studentaccessibility@mountunion.edu. More information on Student Accessibility Services can be located at: <https://www.mountunion.edu/academics/student-success/accessibility>

COUNSELING SERVICES:

The aim of counseling at Mount Union is to assist students in becoming socially and emotionally healthy. Counseling is a partnership between client and counselor, meant to help students find solutions and possibilities in life. In the Office of Counseling Services, state licensed counselors help Mount Union students move toward life goals, while teaching the coping skills needed for the future. Counseling staff members provide free and confidential counseling by appointment to enrolled students. Walk-in hours are also available M, W, F 2-4PM and T, TH 11am-1pm. No appointment is needed for walk-in hours. During these brief meetings, students can initiate services or meet with a counselor to discuss unexpected situations that are creating emotional discomfort. The Department of Counseling Services is located within the Student Affairs Suite in the Hoover-Price Campus Center. To schedule an appointment, call 330.823.2886 or email counselingservices@mountunion.edu.

RAIDERS COVE:

Located in Van Den Eynden Hall, Raiders Cove assists students with basic living needs such as food, clothing, or hygiene resources. There are also Raiders Cove Grab & Go locations in various offices across campus (e.g., the IC Office in KHIC 110A and the MLC director's office in KHIC 258). Visit www.mountunion.edu/raiders-cove for hours and more information.

ADDITIONAL WELLNESS SERVICES:

Students can also access services from the following offices: * Health Center, 330.596.7995, health-center@mountunion.edu * Alcohol, Drug, and Wellness Education, 330.829.6660, adwe@mountunion.edu * University Chaplain, 330-823-8715, chapel@mountunion.edu

Schedule

Week 1 (Week of 1-12)

Monday

- **Lecture:** Introduction to PyTorch
- **Recommended Reading:**
 - *Artificial Intelligence: A Modern Approach* Chapters 19-21, Appendix A
 - [PyTorch's Get Started Guide](#)
 - [PyTorch's Tutorials](#)
 - [PyTorch's Documentation](#)
 - [Python Documentation on Classes and OOP](#)
 - [Python Classes in 1 Minute! \(Video\)](#)
 - [Python OOP Tutorial \(Video Playlist\)](#) ##### Wednesday
- **Lecture:** Review of Recurrent Neural Networks, Transformers, and Large Language Models
- **Recommended Reading:**
 - *Artificial Intelligence: A Modern Approach* Chapter 23-24 ##### Friday
- **Lecture:** Implementing Large Language Models
- **Recommended Reading:**

Week 2 (Week of 1-19)

Monday

- **No Class, MLK Day** #### Wednesday
- **Lecture:** Attention Variants
- **Recommended Reading:** #### Friday
- **Lecture:** Retrieval Augmented Generation (RAG), Explainable Artificial Intelligence
- **Recommended Reading:**

Week 3 (Week of 1-26)

Monday

- **Lecture:** Review of Convolutional Neural Networks
- **Recommended Reading:** #### Wednesday
- **Lecture:** Pretrained CNN Models
- **Recommended Reading:**
- **Homework 1: Natural Language Processing** Due by 11:59pm #### Friday
- **Lecture:** Object Detection and Segmentation
- **Recommended Reading:**
- **Quiz 1:** Natural Language Processing

Week 4 (Week of 2-2)

Monday

- **Lecture:** CNNs and Transformers
- **Recommended Reading:**
 - *Artificial Intelligence: A Modern Approach* Chapter 25 #### Wednesday
- **Lecture:** Diffusion Models
- **Recommended Reading:** #### Friday
- **Lecture:** Diffusion Models
- **Recommended Reading:**

Week 5 (Week of 2-9)

Monday

- **Lecture:** Introduction to Graph Neural Networks (GNNs)
- **Recommended Reading:** ##### Wednesday
- **Lecture:** Graph Attention Networks (GATs)
- **Recommended Reading:**
- **Homework 2: Computer Vision** Due by 11:59pm ##### Friday
- **Lecture:** Applications of GNNs
- **Recommended Reading:**
- **Quiz 2:** Computer Vision

Week 6 (Week of 2-16)

Monday

- **Lecture:** Introduction to Reinforcement Learning
- **Recommended Reading:**
 - *Artificial Intelligence: A Modern Approach* Chapter 22 ##### Wednesday
- **Lecture:** Deep Q Networks
- **Recommended Reading:** ##### Friday
- **Lecture:** Applications of Reinforcement Learning
- **Recommended Reading:**

Week 7 (Week of 2-23)

Monday

- **Lecture:** Introduction to Generative Adversarial Networks (GANs)
- **Recommended Reading:** ##### Wednesday
- **Lecture:** GAN Variants
- **Recommended Reading:**
- **Homework 3: Graph Neural Networks and Reinforcement Learning** due by 11:59pm ##### Friday
- **Lecture:** Applications of GANs
- **Quiz 3:** Graph Neural Networks and Reinforcement Learning

Week 8 (Week of 3-2)

Monday

- **Lecture:** Explainability of AI Revisited #### Wednesday
- **Lecture:** Ethics of Artificial Intelligence
- **Recommended Reading:**
 - *Artificial Intelligence: A Modern Approach* Chapter 27-28
- **Quiz 4:** Generative Artificial Intelligence
- **Homework 4: Generative Artificial Intelligence** due by 11:59pm #### Thursday
- **Final Exam Period**
- Final Project Presentations